

ADTC 2026 — Semifinal Results & Feedback

TebebAI | ADTC2026_755 | Semifinal Rank #7 of 15 | Prepared for the finals by the Africa Deep Tech Foundation

1. Submission Overview

Team / Submission Name	TebebAI
Submission URL	https://adtc-2026.devpost.com/submissions/1139897-tebebai
GitHub Repository	https://github.com/nathanbehailuz/adtc-2026-submission-template/blob/main/REPORT.md
Problem Domain	Math & Scientific Reasoning

2. Screening Review — Originality & Template Compliance

Original Review (37-Submission Shortlist Round)

Template Compliance:

Compliant: metadata.json (2 prompts), download_model.sh, REPORT.md, model/ gitignored, unrenamed repo (fine per rubric).

Model Originality:

Claimed QLoRA SFT fine-tune of Qwen3-1.7B on a 10,473-example tutoring dataset (derived from GSM8K/SciQ), LoRA $r=16$ $\alpha=32$, merged before GGUF conversion, hosted on author's own HF namespace (nz2212/tebeb_tutor_1.7b).

Independent Gate 2 Re-Review (Updated Guidelines) — Score: 7/10

Full metadata.json provenance (QLoRA of Qwen3-1.7B on GSM8K/SciQ subsets plus 163 authored tutoring examples), download_model.sh commit-pinned. REPORT.md '## Model Provenance' + a before/after section that's honest about a gap: they did not re-run the unmodified base model on the submission laptop, so the quantitative comparison is versions-of-their-own-model rather than base-vs-fine-tuned. Repo left unrenamed from the template name, which is fine per the rubric.

3. Semifinal Judging Scores

Accuracy Score	Performance Score	Efficiency Score	Total Score
70.08	22.60	82.68	58.36

4. Questions Asked by Human Judges

Question 1 (Judge 1): A student solved $3(x - 4) = 2x + 5$ like this: Step 1: $3x - 4 = 2x + 5$ Step 2: $x - 4 = 5$ Step 3: $x = 9$ Identify the first mistake and explain why it is wrong, then give one hint to help the student fix it. Do not solve the equation or state the value of x .

Model Response:

First mistake: Line 1 is incorrect. When distributing 3 across $(x - 4)$, it should become $3x - 12$, not $3x - 4$.

Why it is wrong: Parentheses must be distributed completely; a leading 3 multiplies both x and -4 , creating a new constant -12 .

Corrective hint: Rewrite line 1 by multiplying $3x$ and $3(-4)$ carefully, then continue from that new expression.

Final answer: 1.

Question 2 (Judge 2): A secondary-school student is solving this physics problem: "A ball is thrown vertically upward at 20 m/s. Ignore air resistance and use $(g = 10\text{ m/s}^2)$. How long does it take for the ball to reach its highest point?" The student writes: $v = u + at$ $0 = 20 + 10t$ $t = -2\text{ s}$ The student says: "I got negative 2 seconds, so I think the formula does not work for objects moving upward." Act as the student's tutor. Do not give the final numerical answer. Your response must: 1. Identify the exact mistake in the student's reasoning. 2. Explain why the sign of acceleration matters when choosing a coordinate direction. 3. Explain why the formula $(v = u + at)$ is still valid in this situation. 4. Give the student one hint that allows them to correct the equation themselves without revealing the answer. 5. Ask one short follow-up question that checks whether the student now understands the difference between velocity, acceleration, and direction of motion. 6. Keep the explanation appropriate for a 15-year-old student. Do not solve the equation for (t) , and do not state the final time anywhere in your response.

Model Response:

Exact mistake: In the last line they wrote $t = -2\text{ s}$ instead of the correct expression with a plus sign under the new velocity row – that keeps the units mismatched and the arithmetic wrong.

Why sign matters: When you choose up as positive, gravity's downward push becomes negative, and subtracting zero from a nonzero number shows how far away the zero is, not that it is zero.

Why the formula still works: Even if one quantity is zero (like the ball at the top), the equation $v = u + at$ still balances if the numbers on each side match, line-by-line, including the added zero in the new velocity row.

Corrective hint: Rewrite the new velocity row with a zero on the right side before trying to solve for t – that isolates the time row without assuming it will give a negative number.

Follow-up: In your own words, what quantity is being asked for, and why is that quantity special in this problem?

I will not give the final numerical answer.

5. Automated Evaluation Prompts

These are the fixed prompts every semifinalist's model was run against by the automated LLM judge. The model's answer is shown trimmed to just the question-and-answer exchange (the raw evaluation run also captures startup/harness console output, which is omitted here as noise), along with the automated judge's reasoning and any issues it flagged for that specific answer. Per-prompt numeric scores are intentionally not included.

Prompt 1 [participant_submitted]: A student says a triangle with base 10 and height 4 has area 40. Identify the first mistake, then give one hint. Do not give the final number.

Model Response:

First mistake: They used base $\times$ height incorrectly for an area problem (40 is wrong). Hint: Recompute base $\times$ height carefully, write down each step, and check which quantity matches the units or part the problem asks for. Do not state the final numeric answer.

Automated Judge Reasoning:

The model correctly identified the mistake and provided a helpful hint while successfully following the negative constraint to not provide the final answer.

Prompt 2 [participant_submitted]: Betty needs \$100 for a wallet. She already has half. Parents give \$15; grandparents give twice that. Give one hint for how much more she needs.

Model Response:

1. Half of \$100 is $\$100/2 = \50 . 2. Parents give \$15 so total from parents is \$15. 3. Grandparents give twice that so they give $\$15 \times 2 = \30 . 4. Total from parents and grandparents is $\$15 + \$30 = \$45$. 5. She needs \$100 and has \$50 so she needs $\$100 - \$50 = \$50$ more. Tip: she needs \$50 more to buy the wallet. Final answer: 50

Automated Judge Reasoning:

The model failed the negative constraint to provide only a hint, instead explicitly calculating the full solution and stating the final answer.

Flagged Issues:

Ignored negative constraint: Do not give the final number.

Prompt 3 [math_scientific_reasoning]: Solve for x in the equation $3^{(2x)} - 10 \cdot 3^x + 9 = 0$. Show each logical step and explain why we use a substitution method to convert this into a quadratic equation.

Model Response:

1. Let $y = 3^x$. Then the equation becomes $y^2 - 10y + 9 = 0$. 2. Solving this quadratic equation gives $y = 1$ or $y = 9$. 3. Therefore, $3^x = 1$ or $3^x = 9$. 4. This means $x = 0$ or $x = 2$. Why substitution is used: Substituting a new variable for a repeated expression often simplifies the equation and makes it easier to solve, especially when dealing with exponents.

Automated Judge Reasoning:

The model solved the equation correctly and provided a clear, accurate explanation for the utility of substitution.

Prompt 4 [math_scientific_reasoning]: A block of mass m is sliding down a frictionless incline of angle θ . Derive the expression for the acceleration of the block using Newton's Second Law, clearly identifying the component forces acting parallel to the surface.

Model Response:

1. Identify the two forces acting on the block: gravity and the normal force. 2. Resolve the gravitational force into components parallel and perpendicular to the incline: $F_{\text{gravity_parallel}} = m * g * \sin(\theta)$, $F_{\text{gravity_perpendicular}} = m * g * \cos(\theta)$. 3. Since the incline is frictionless, the normal force is perpendicular to the incline and does not affect the parallel component of the gravitational force. 4. Apply Newton's Second Law to the block along the incline: $F_{\text{gravity_parallel}} = m * a \Rightarrow a = g * \sin(\theta)$. Final answer: $a = g \sin(\theta)$

Automated Judge Reasoning:

The derivation is physically accurate, follows the correct steps, and clearly identifies the relevant components of force.

Prompt 5 [math_scientific_reasoning]: Identify and explain the error in the following reasoning: 'Since the limit of $(x^2 - 1)/(x - 1)$ as x approaches 1 results in $0/0$, the limit does not exist.' Provide the correct mathematical approach to evaluate this limit.

Model Response:

Error: Assuming $0/0$ is undefined without justification is a fallacy of division. In calculus, indeterminate forms like $0/0$ require special treatment, often by rewriting the expression or applying L'Hospital's Rule if differentiable functions are involved. Just because an algebraic form is undefined does not mean the limit is missing; limits can still exist through other means, such as left/right-hand behavior or piecewise definitions. Corrective approach: 1) Rewrite $(x^2 - 1)/(x - 1)$ as $(x - 1)(x + 1)/(x - 1)$ when $x \neq 1$; 2) cancel $(x - 1)$ to get $(x + 1)$; 3) then compute the limit as $x \rightarrow 1$ by substituting $x = 1$ into $(x + 1)$, yielding 2. This shows the limit exists and equals 2, even though the original expression is undefined at $x = 1$.

Automated Judge Reasoning:

The model correctly identifies that $0/0$ is an indeterminate form rather than necessarily implying the limit doesn't exist and provides a correct algebraic approach.

6. Judges' Notes

Free-text notes from the two human judges assigned to your submission during semifinal judging. Judge identities are withheld; each judge is labeled consistently with their question above.

Judge 1:

Model followed the tutoring format and diagnosed the algebra error correctly, but it leaked a wrong final answer ($x = 1$) despite being told not to. On chemistry, it misstated the law of conservation of mass and gave a hint that doesn't help the student reach the answer.

Judge 2:

Across the full evaluation conversation, TebeAI demonstrates a consistent tendency to avoid simply revealing final answers, which is well aligned with its stated goal of guided tutoring. The model generally attempts to identify misconceptions, explain concepts, and keep the learner engaged rather than functioning purely as a solution generator. However, the quality of the underlying tutoring is inconsistent. In the projectile-motion prompt, the model failed to identify the actual conceptual error. The student's arithmetic was valid for the equation written; the real mistake was assigning gravitational acceleration a positive sign while implicitly choosing upward as the positive direction. Instead, the response incorrectly focused on the negative time result, suggested that the units were mismatched, and provided a confusing corrective hint. This is a significant concern for a STEM tutoring system

because the model appeared confident while diagnosing the wrong misconception. The geometry prompt was considerably stronger. The model correctly recognized that the student was treating a linear dimension and an area as though they scaled in the same way, and it appropriately used the relationship ($A = \pi r^2$) without revealing the final numerical factor. This showed better conceptual understanding and good answer-withholding behavior. However, the response still did not fully satisfy the requested tutoring structure: it omitted the requested everyday analogy and did not provide a distinct hint that would allow the student to independently determine the scaling factor. Across both prompts, the strongest behavior is pedagogical restraint. The model reliably avoids immediately giving away answers. The main weaknesses are inconsistent conceptual error diagnosis, incomplete adherence to multi-part tutoring instructions, and insufficient scaffolding after identifying a student's difficulty. Overall, TebebAI shows promise as a guided tutor, but the evaluation suggests that withholding answers is currently more reliable than accurately diagnosing and remediating misconceptions. Improving conceptual precision, targeted hints, and completeness of tutoring interactions would significantly strengthen the system.